RxGuard: Knowledge-Guided Safety Guardrails for Medication Recommendation

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Abstract

Medication combination recommendation predicts a set of medications for a patient visit from longitudinal electronic health records (EHRs) and is safety-critical, since feasibility depends on set-level constraints such as drug–drug interactions (DDIs). Despite substantial progress, existing approaches suffer from three limitations: (1) safety is often imposed as a training-time soft regularizer, while inference relies on thresholding or top-k selection over independent drug scores, providing no executable per-instance feasibility control; (2) external knowledge is mainly used for representation enrichment rather than being compiled into inference-ready decision artifacts that directly support constraint execution and evidence retrieval; and (3) explanations are typically inclusion-centric, lacking structured, auditable rationales for constraint-triggered exclusions. To address these gaps, we propose **RxGuard** (Rx denotes prescription), a novel decision-oriented framework that formulates medication combination recommendation as utility maximization under explicit feasibility constraints. Technically, RxGuard learns trajectory-aware medication utilities and a candidate set from EHRs, compiles external knowledge into an executable feasibility interface and an Evidence KG for audit retrieval, and performs guarded set selection at inference time to enforce feasibility by construction while producing decision-consistent audit records with grounded rationales for both inclusions and exclusions. Experiments on MIMIC-III and MIMIC-IV demonstrate that RxGuard improves both recommendation performance and safety, while providing consistent, decision-aligned auditing.

CCS Concepts

• **Applied computing** → **Health informatics**; • **Computing methodologies** → **Knowledge representation and reasoning**; **Machine learning**.

Keywords

medication recommendation, medication combination recommendation, electronic health records, knowledge graphs, drug-drug interactions

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1 Introduction

Medication combination recommendation aims to predict an appropriate medication set for a patient visit from electronic health records (EHRs) and is a key task in clinical decision support [1, 16, 30, 40]. Unlike conventional multi-label prediction, this task depends not only on therapeutic relevance but also on set-level safety feasibility: the recommended set must meet interaction-safety feasibility requirements, such as avoiding harmful drug–drug interactions (DDIs) [3–5, 16, 30–32, 34, 38]. Since inappropriate combinations may lead to severe adverse events, clinical deployment requires not only predictive performance but also instance-level, verifiable safety control and evidence to support clinical review [6–9, 12, 15, 27, 33].

Prior research has mainly evolved along two technical directions. One line of work learns patient representations from longitudinal visit trajectories using sequential modeling and attention mechanisms to improve predictive performance [4, 5, 16, 31]. The other line incorporates external knowledge resources, such as interaction networks, prescription relation structures, molecular graphs, and multi-relational knowledge graphs (KGs), to enhance representations, inject safety signals, and provide supporting evidence or traceability [3, 14, 17–20, 22–25, 28–30, 32, 34, 35, 37–42]. While these approaches have improved representations and knowledge utilization, their inference pipelines typically rely on independent drug scoring followed by heuristic filtering, which limits their ability to meet the clinical requirements of set-level safety control and process auditability.

Specifically, existing methods exhibit three limitations: **(1) Lack of executable set-level safety control at inference time.** Many safety-aware models introduce interaction risks as soft constraints during training, whereas the final prescription is commonly constructed by thresholding or top-k selection over independent drug scores; this strategy does not explicitly execute feasibility constraints on the generated set and therefore cannot provide verifiable instance-level safety control. **(2) External knowledge is not organized into inference-ready decision structures.** External knowledge is often used implicitly to influence representations and scores. Even when graph-structured knowledge is utilized during training, inference typically yields only scores or probabilities, which are

insufficient to directly support constraint execution and evidence retrieval. **(3) Explanations are misaligned with set construction, especially for exclusions.** When providing evidence, existing approaches usually focus on justifying included medications, while exclusion decisions triggered by interaction risks or other set-level constraints are rarely documented in a decision-consistent manner.

To address these limitations, we propose **RxGuard**, a decision-oriented framework for safe medication recommendation. Here, Rx is a common abbreviation for prescription, and RxGuard denotes introducing explicit safeguards into the prescription generation process. Unlike prior approaches that primarily suppress risks during training and rely on post-hoc filtering at inference, RxGuard shifts safety control to the decision stage by transforming external knowledge from implicit representational signals into inference-ready decision artifacts that simultaneously support verifiable safety enforcement and auditable evidence recording. As illustrated in Figure 1, RxGuard consists of three interconnected modules. First, **Trajectory-Aware Medication Scoring** (Sec. 3.2) models longitudinal EHR trajectories to estimate medication utilities and construct a compact candidate medication set for the target visit. Second, **Decision KG Compilation** (Sec. 3.3) conditions on the visit context and the candidate set to compile external knowledge into two complementary visit-specific KGs: a Constraint KG that is further compiled into an executable feasibility interface for set-level safety checking, and an Evidence KG that supports retrievable, structured evidence for subsequent audit, thereby making knowledge directly usable during inference rather than only influencing internal representations. Finally, **Guarded Set Selection** (Sec. 3.4) explicitly executes the compiled feasibility interface during set construction to generate a constraint-satisfying recommendation and produces a decision-consistent audit record with inclusion evidence and constraint-aligned exclusion rationales to support clinical review.

Our contributions are summarized as follows:

- We propose RxGuard, a novel framework that treats safe medication recommendation as an explicit decision-constrained task. Unlike prior methods that rely on training-time risk suppression and post-hoc filtering, RxGuard shifts safety control to the inference stage by executing compiled feasibility constraints, enabling verifiable, instance-level set safety.
- We introduce a knowledge compilation mechanism that transforms external knowledge into inference-ready decision artifacts. RxGuard constructs a visit-specific Constraint KG and compiles it into an executable feasibility interface for safety enforcement, while building a visit-specific Evidence KG for retrievable traceability without affecting the recommendation outcome, so that knowledge supports enforcement and review beyond implicit representation learning.
- Experiments on real-world EHR datasets show that RxGuard improves both effectiveness and safety over strong baselines, with additional analyses further validating the necessity of each module.

2 Related Work

This section reviews prior work on medication recommendation from EHRs and external medical knowledge. We categorize existing methods into three groups according to the form of external knowledge used: (1) EHR-based medication recommendation that relies solely on longitudinal EHR signals, (2) medication recommendation with structured drug knowledge that augments EHR models with non-KG structured resources, and (3) KG-enhanced medication recommendation that leverages multi-relational KGs for recommendation and evidence support.

2.1 EHR-based Medication Recommendation

Prior work on EHR-based approaches commonly formulates medication recommendation as a multi-label prediction problem over longitudinal EHRs, where the input is a patient's visit history and the output is the medication set for a target visit. The central goal is to learn patient-state representations from sequential visit data and directly decode a set of medications. Doctor AI [4] models visit sequences with recurrent neural networks and performs end-to-end prediction of subsequent clinical events, including medication-related outcomes. RETAIN [5] introduces reverse-time attention to aggregate historical visits, enabling both competitive prediction and a visit-level attribution that supports interpretability. TAHD-Net [31] incorporates time information and hierarchical dependency modeling to better capture long-range dependencies and structured relations among clinical events for medication set prediction. DMNC [16] proposes a dual-memory architecture for asynchronous multi-view sequences; although not designed exclusively for medication recommendation, its memory-based fusion mechanism is frequently adopted to integrate heterogeneous clinical signals and support medication prediction. Overall, these EHR-only approaches establish the basic paradigm of learning from longitudinal EHR trajectories, but they largely rely on endogenous EHR signals and thus have limited capacity to systematically encode medication interactions and pharmacological properties that are crucial for combination safety.

2.2 Medication Recommendation with Structured Drug Knowledge

To address set-level safety risks, many studies enhance EHR models with structured drug knowledge, including DDI graphs, prescription relation networks, and molecular structure information. In most cases, these resources are used to learn drug representations or introduce safety-aware objectives during training, rather than to produce executable constraints at decision time. GAMENet [30] integrates drug relation graphs through a memory-augmented architecture and employs interaction-aware learning to reduce unsafe co-prescriptions. SafeDrug [38] and CARMEN [3] incorporate molecular structure information, together with interaction signals, to improve safety-aware set prediction, while MoleRec [39] further strengthens molecular modeling with substructure-aware representations. Several methods emphasize longitudinal patient modeling with explicit risk control, including DrugRec [32] and 4SDrug [34], the latter additionally encouraging smaller yet safer medication sets. DGCL [18] improves robustness through contrastive learning over drug graphs, and Trans-GAHNet [17] introduces multi-view graph

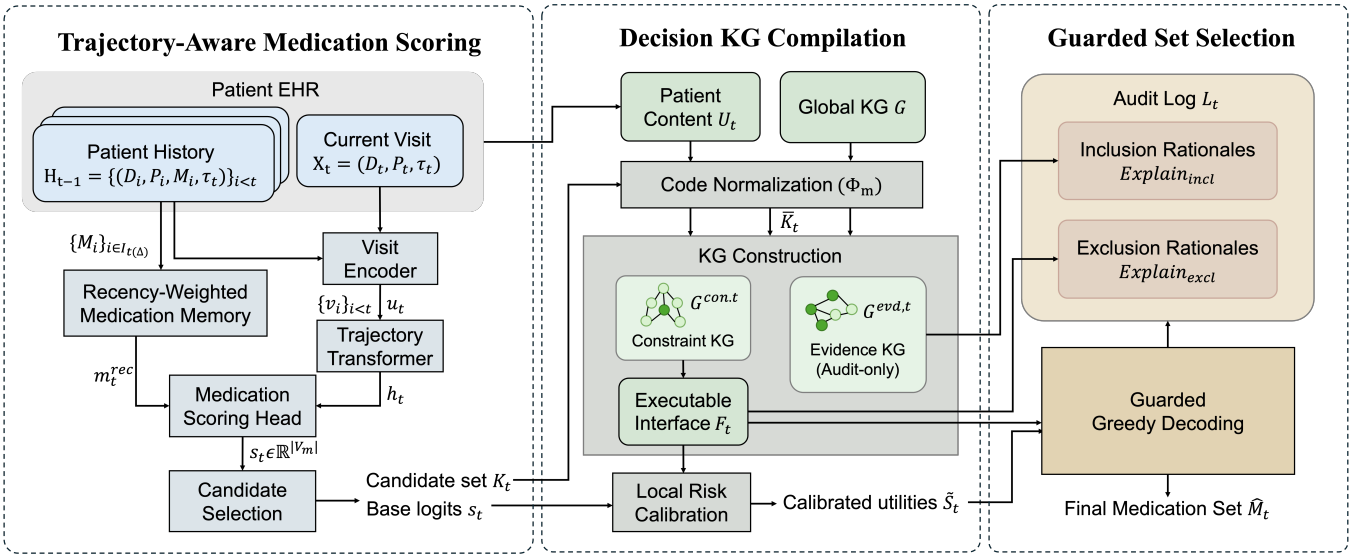


Figure 1: Overview of RxGuard. RxGuard couples patient-side EHR modeling with knowledge-side decision compilation via inference-ready structures. (1) Trajectory-Aware Medication Scoring (Section 3.2) encodes longitudinal EHR trajectories to output base logits s_t and a candidate set K_t. (2) Decision KG Compilation (Section 3.3) uses the patient context $\mathcal{U}_t$, the normalized candidate set $\bar{K}_t = \Phi_m(K_t)$, and external knowledge (Global KG G) to construct a Constraint KG $G_{\text{con},t}$ and an Evidence KG $G_{\text{evd},t}$, compile an executable feasibility interface F_t (forbidden medication pairs), and perform local risk calibration on the constraint-induced candidate graph to obtain KG-calibrated utilities $\tilde{s}_t$. (3) Guarded Medication Set Selection (Section 3.4) runs guarded greedy decoding over $\bar{K}_t$ using $\tilde{s}_t(\cdot)$ while explicitly enforcing F_t, producing a feasible medication set $\hat{M}_t \subseteq \bar{K}_t$; inclusion rationales are retrieved from $G_{\text{evd},t}$, and decision-consistent exclusion rationales are derived from F_t.

augmentation to inject structured drug signals into Transformer-based patient encoders. AKA-SafeMed [40] adopts knowledge augmentation over external drug relation structures to reinforce safety awareness, while DrugDoctor [14] addresses cold-start scenarios through representation learning with safety-oriented designs. Recent work further explores multi-view fusion and label-aware learning, including MGRN [23] and LAMRec [35]. Overall, these approaches substantially advance the use of interaction and structural signals for safer recommendations. However, external knowledge is typically incorporated as single-relation networks or auxiliary structural views for representation learning and training-time regularization, which rarely yields an executable, instance-specific feasibility interface for decision-time constraint enforcement or decision-consistent structured evidence suitable for clinical auditing.

2.3 KG-enhanced Medication Recommendation

KG-enhanced approaches organize clinical and pharmacological knowledge as multi-entity, multi-relation KGs, providing a structured substrate for evidence aggregation and graph-based modeling beyond single-network interaction representations. MedRe [42] combines EHR signals with KG information via KG-enhanced attributed multi-task learning to mitigate sparsity and improve recommendation performance. KGDNet [25] jointly models longitudinal EHR signals and KG structural information to improve effectiveness and clinical plausibility. SSPNet [41] integrates external knowledge to enhance robustness under noisy or shifted clinical contexts. PROMISE [37] leverages knowledge infusion with pretraining-based representation learning to improve generalization. TraceDR [20] emphasizes traceable recommendation over a medical KG and produces explicit graph evidence to strengthen auditability. Overall, KG-enhanced approaches provide a principled way to introduce structured medical knowledge; however, most existing methods primarily use KGs for representation augmentation, while offering limited mechanisms to transform KG information into executable safety constraints at decision time and to output decision-consistent structured evidence that supports clinical review.

3 Method

This section presents the RxGuard framework, as shown in Figure 1. We first introduce the problem setup and notation, and then describe three modules: (1) *Trajectory-Aware Medication Scoring* models the longitudinal EHR to output base logits s_t and a candidate set K_t; (2) *Decision KG Compilation* takes the patient context $\mathcal{U}_t$, the candidate set K_t (after code normalization), and external knowledge represented by a global KG G to induce a Constraint KG $G_{\text{con},t}$ and an Evidence KG $G_{\text{evd},t}$, compile an executable feasibility interface F_t, and perform local risk calibration on the constraint-induced candidate graph to obtain KG-calibrated utilities $\tilde{s}_t$; (3) *Guarded Set Selection* explicitly executes F_t during guarded greedy decoding to produce a feasible medication set $\hat{M}_t$ and structured audit rationales, including inclusion evidence retrieved from $G_{\text{evd},t}$ and decision-consistent exclusion rationales aligned with F_t.

3.1 Problem Setup and Notation

EHR Data. Let $\mathcal{P}$ denote the patient set. For a patient $p \in \mathcal{P}$, the longitudinal EHR is represented as an ordered visit sequence

$$\mathbf{X}^{(p)} = \left[x_1^{(p)}, x_2^{(p)}, \ldots, x_{T_p}^{(p)}\right], \tag{1}$$

where T_p is the number of visits. Each visit is recorded as a tuple of set-structured clinical concepts with a timestamp

$$x_t^{(p)} = \left(D_t^{(p)},\ P_t^{(p)},\ M_t^{(p)},\ \tau_t^{(p)}\right), \tag{2}$$

where $D_t^{(p)} \subseteq \mathcal{V}_d$ is the set of diagnoses, $P_t^{(p)} \subseteq \mathcal{V}_p$ is the set of procedures, and $M_t^{(p)} \subseteq \mathcal{V}_m$ is the medication set prescribed at visit t; $\tau_t^{(p)}$ denotes the visit timestamp. Here, $\mathcal{V}_d$, $\mathcal{V}_p$, and $\mathcal{V}_m$ denote the vocabularies of diagnoses, procedures, and medications, respectively. For clarity, when focusing on a single patient, the superscript (p) is omitted.

At a target visit t, the observable current visit context and the available history are defined as

$$X_t = (D_t, P_t, \tau_t), \tag{3}$$

$$H_{t-1} = \left[(D_i, P_i, M_i, \tau_i)\right]_{i=1}^{t-1}. \tag{4}$$

The set-valued patient context at visit t is defined as

$$\mathcal{U}_t = D_t \cup P_t, \tag{5}$$

which will be used for knowledge conditioning and decision KG compilation.

Concept Normalization. To align EHR codes with the external KG, diagnosis and procedure codes are standardized in preprocessing by mapping them into the UMLS concept identifier (CUI) space using [21]. Accordingly, D_t and P_t in this section refer to the standardized CUIs, and $\mathcal{U}_t = D_t \cup P_t$ denotes the standardized visit context set. In particular, $\mathcal{V}_d$ and $\mathcal{V}_p$ are treated as the corresponding vocabularies in the standardized CUI space throughout this section.

To align EHR medication identifiers with external knowledge sources, a normalization map Φ_m converts raw medication identifiers to a fixed canonical identifier space:

$$\bar{d} = \Phi_m(d), \qquad \bar{K}_t = \Phi_m(K_t), \tag{6}$$

where $d \in \mathcal{V}_m$ and $K_t \subseteq \mathcal{V}_m$ denotes the candidate medication set at visit t.

Medication Set Prediction. Given the longitudinal history and the observable information at the target visit, the goal is to predict the medication set prescribed at that visit. Specifically, the objective is to learn a parameterized predictor

$$f_\theta : (H_{t-1}, X_t) \to \hat{M}_t \subseteq \mathcal{V}_m. \tag{7}$$

3.2 Trajectory-Aware Medication Scoring

Given the available history H_{t-1} and the observable current visit context X_t (defined in Section 3.1), this module outputs base medication logits $s_t \in \mathbb{R}^{|\mathcal{V}_m|}$ and a candidate set $K_t \subseteq \mathcal{V}_m$.

Visit Encoder. Each clinical concept c (diagnosis, procedure, or medication) is associated with a learnable embedding $e_c \in \mathbb{R}^d$. For any set of concepts S, a set representation is obtained via mean pooling:

$$\mathrm{Pool}(S) = \frac{1}{|S|} \sum_{c \in S} e_c. \tag{8}$$

If $S = \emptyset$, $\mathrm{Pool}(S) = \mathbf{0}$ is used. The target-visit representation $u_t \in \mathbb{R}^{2d}$ is formed by concatenating pooled diagnosis and procedure embeddings, while each historical-visit representation $v_i \in \mathbb{R}^{3d}$ further concatenates the pooled medication embedding from that visit.

Trajectory Transformer. To model long-range dependencies under irregular visit intervals, the historical visits are treated as a token sequence augmented with time-gap information. Let $\Delta_{t,i} = \tau_t - \tau_i \geq 0$ denote the time gap between a historical visit i and the target visit t. A learnable time-gap embedding function $\phi(\cdot)$ maps the scalar gap to $\mathbb{R}^{d_h}$. Each token is formed by combining a projected visit representation with a time-gap embedding:

$$z_i^{(0)} = W_p v_i + \phi(\Delta_{t,i}), \tag{9}$$

where $W_p \in \mathbb{R}^{d_h \times 3d}$. A L-layer Transformer encoder contextualizes the token sequence and yields $\{z_i^{(L)}\}_{i<t}$. A target-conditioned readout yields the trajectory state $h_t \in \mathbb{R}^{d_h}$:

$$\alpha_{t,i} = \mathrm{softmax}_i\left(\frac{(W_q u_t)^\top (W_k z_i^{(L)})}{\sqrt{d_h}}\right), \qquad h_t = \sum_{i<t} \alpha_{t,i}\,(W_v' z_i^{(L)}), \tag{10}$$

where $W_q \in \mathbb{R}^{d_h \times 2d}$, $W_k \in \mathbb{R}^{d_h \times d_h}$, and $W_v' \in \mathbb{R}^{d_h \times d_h}$ are learnable parameters.

Recency-Weighted Medication Memory. To capture short-term regimen continuity, medications from recent visits are aggregated with time decay. Let $\Delta > 0$ denote the recency window size and define the recent-visit index set $\mathcal{I}_t(\Delta) = \{\, i < t \mid \tau_t - \tau_i < \Delta \,\}$. If $\mathcal{I}_t(\Delta) = \emptyset$, $m_t^{rec} = \mathbf{0}$ is used. Otherwise, the recency-weighted medication memory is computed as a time-decayed weighted average of pooled medication embeddings:

$$m_t^{rec} = \sum_{i \in \mathcal{I}_t(\Delta)} \tilde{w}_{t,i}\,\mathrm{Pool}(M_i) \in \mathbb{R}^d, \tag{11}$$

$$\tilde{w}_{t,i} = \frac{\exp\big(-\gamma(\tau_t - \tau_i)\big)}{\sum_{j \in \mathcal{I}_t(\Delta)} \exp\big(-\gamma(\tau_t - \tau_j)\big)},$$

where $\gamma > 0$ controls the decay rate.

Medication Scoring and Candidate Selection. The trajectory state h_t and the recency memory m_t^{rec} are fused to produce base medication logits:

$$s_t = W_s\,[h_t;\, W_m m_t^{rec}] + b_s, \tag{12}$$

where $W_m \in \mathbb{R}^{d_h \times d}$, $W_s \in \mathbb{R}^{|\mathcal{V}_m| \times 2d_h}$, and $b_s \in \mathbb{R}^{|\mathcal{V}_m|}$ are learnable parameters. Based on s_t, a candidate set of size N is obtained by top-N selection:

$$K_t = \mathrm{TopN}(s_t), \qquad |K_t| = N, \tag{13}$$

where $\mathrm{TopN}(\cdot)$ returns the indices of the N largest entries in s_t. During training, let $y_t \in \{0,1\}^{|\mathcal{V}_m|}$ be the multi-hot label vector

indicating the ground-truth medication set at visit t; the scorer is optimized with the multi-label logistic loss:

$$\mathcal{L}_{\text{pred}} = -\sum_{t}\sum_{j=1}^{|\mathcal{V}_m|}\Big(y_{t,j}\log\sigma(s_{t,j})+(1-y_{t,j})\log(1-\sigma(s_{t,j}))\Big), \quad (14)$$

where $\sigma(\cdot)$ is the sigmoid function.

3.3 Decision KG Compilation

Given the patient context $\mathcal{U}_t$ and the candidate set K_t from Section 3.2, together with external knowledge represented by a Global KG G, this module compiles visit-specific decision structures for safety control, score calibration, and subsequent audit retrieval. Specifically, it (i) induces a Constraint KG $G_{\text{con},t}$ and compiles an executable feasibility interface F_t, (ii) induces an Evidence KG $G_{\text{evd},t}$ for audit retrieval (without affecting the recommendation outcome), and (iii) performs local risk calibration to produce KG-calibrated utilities $\tilde{s}_t$.

Global KG.. External knowledge is represented as a typed multi-relational knowledge graph $G = (V, E, \mathcal{R})$, where each edge is a triple $(u, r, v) \in V \times \mathcal{R} \times V$ with relation type $r \in \mathcal{R}$. Relation types are partitioned into constraint relations $\mathcal{R}_{\text{con}}$ (DDI constraints between medications) and evidence relations $\mathcal{R}_{\text{evd}}$ (clinically relevant associations used for audit).

Constraint KG and Executable Interface. The canonical candidate set is obtained as $\bar{K}_t = \Phi_m(K_t)$ after code normalization defined in Section 3.1. For the current visit, the decision-relevant node set is

$$V_t = \bar{K}_t \cup \mathcal{U}_t. \quad (15)$$

The Constraint KG is induced by selecting constraint edges among candidate medications:

$$E_{\text{con},t} = \{(x, r, y) \in E \mid x \in \bar{K}_t,\ y \in \bar{K}_t,\ r \in \mathcal{R}_{\text{con}}\}, \quad (16)$$

and $G_{\text{con},t} = (V_t, E_{\text{con},t})$. From $G_{\text{con},t}$, an executable feasibility interface is compiled as a set of forbidden unordered pairs:

$$F_t = \Big\{\{\bar{a}, \bar{b}\} \subseteq \bar{K}_t \;\Big|\; \bar{a} \neq \bar{b},\ \exists r \in \mathcal{R}_{\text{con}},\ (\bar{a}, r, \bar{b}) \in E_{\text{con},t}\Big\}. \quad (17)$$

In implementation, constraints are handled as unordered pairs, and edge direction is ignored when compiling F_t for feasibility checking.

Evidence KG for Audit Retrieval. The Evidence KG is constructed at each visit to provide structured and traceable evidence for subsequent audit and rationale retrieval, without participating in score calibration or feasibility enforcement. The decision-relevant node scope follows Eq. (15), where $\bar{K}_t$ denotes the canonical candidate medication set and $\mathcal{U}_t$ denotes the current visit context (including diagnoses and procedures). Context-aware edges are selected from the global KG by retaining edges incident to candidate medications whose relation types belong to either constraint or evidence categories:

$$E_{\text{evd},t} = \{(x, r, y) \in E \mid x \in \bar{K}_t,\ y \in V_t,\ r \in \mathcal{R}_{\text{con}} \cup \mathcal{R}_{\text{evd}}\}. \quad (18)$$

To ensure direction-invariant neighbor retrieval, an undirected closure is constructed as

$$E^{\leftrightarrow}_{\text{evd},t} = E_{\text{evd},t} \ \cup\ \{(y, r, x) \mid (x, r, y) \in E_{\text{evd},t}\}, \quad (19)$$

and the Evidence KG is defined as $G_{\text{evd},t} = (V_t, E^{\leftrightarrow}_{\text{evd},t})$. This closure is used solely for retrieval convenience and does not perform semantic inference over relation meanings. The resulting Evidence KG serves exclusively as a knowledge source for audit logging and rationale recording, and does not influence the final recommendation outcome.

Local Risk Calibration and Safety-Aware Training. Local risk calibration injects constraint KG structure into candidate utilities. Canonical base logits on $\bar{K}_t$ are inherited from the scorer under Φ_m:

$$\bar{s}_t(\bar{a}) = \max\{\, s_t(a) \mid a \in K_t,\ \Phi_m(a) = \bar{a}\,\}, \qquad \bar{a} \in \bar{K}_t. \quad (20)$$

Let $\mathcal{N}_t(\bar{a})$ denote the neighbor set of $\bar{a}$ in the constraint-induced candidate graph (i.e., neighbors connected by $E_{\text{con},t}$). The structure-induced conflict risk is defined as the potential co-prescription strength with high-scoring conflicting neighbors:

$$r_t(\bar{a}) = \sum_{\bar{b} \in \mathcal{N}_t(\bar{a})} \sigma(\bar{s}_t(\bar{b})), \qquad \bar{a} \in \bar{K}_t, \quad (21)$$

where $\sigma(\cdot)$ is the sigmoid function. KG-calibrated utilities are then defined as

$$\tilde{s}_t(\bar{a}) = \bar{s}_t(\bar{a}) - \beta\, r_t(\bar{a}), \qquad \bar{a} \in \bar{K}_t, \quad (22)$$

where $\beta > 0$ controls the inhibition strength. The calibrated utilities $\tilde{s}_t$ are used for subsequent set selection, while feasibility is enforced by the compiled interface F_t.

To encourage safety-aware ranking during training, an expected DDI penalty is computed on the compiled interface F_t:

$$\mathcal{L}_{\text{ddi}}(t) = \sum_{\{\bar{a}, \bar{b}\} \in F_t} \sigma(\tilde{s}_t(\bar{a})) \cdot \sigma(\tilde{s}_t(\bar{b})). \quad (23)$$

The overall objective is

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda \sum_{t} \mathcal{L}_{\text{ddi}}(t), \quad (24)$$

where $\lambda \geq 0$ balances effectiveness and safety. The penalty is computed only over the compiled candidate interface F_t, yielding complexity $O(|F_t|)$ per visit. The candidate selection $K_t = \text{TopN}(s_t)$ is treated as a stop-gradient operator for defining the compilation scope.

3.4 Guarded Set Selection

Given the normalized candidate set $\bar{K}_t$, the KG-calibrated utilities $\tilde{s}_t(\cdot)$, the executable feasibility interface F_t, and the Evidence KG $G_{\text{evd},t}$, this module explicitly enforces set-level safety constraints at inference time and produces a feasible medication set $\hat{M}_t$ together with decision-consistent audit rationales.

Guarded Greedy Decoding. Set selection is formulated as a constrained utility maximization problem

$$\begin{aligned} \hat{M}_t &= \arg\max_{M \subseteq \bar{K}_t} \sum_{\bar{a} \in M} \tilde{s}_t(\bar{a}) \\ \text{s.t.} \quad & \{\bar{a}, \bar{b}\} \notin F_t,\ \forall \bar{a} \neq \bar{b} \in M, \quad |M| \leq K_{\max}, \end{aligned} \quad (25)$$

where $K_{\max}$ denotes the maximum medication set size. For computational efficiency, the problem is approximated via greedy feasible decoding. Starting from $S_t = \emptyset$, candidates in $\bar{K}_t$ are sorted in

descending order of $\tilde{s}_t(\bar{a})$. A candidate $\bar{a}$ is admitted if and only if it preserves feasibility with respect to the current set,

$$\forall \bar{b} \in S_t,\ \{\bar{a}, \bar{b}\} \notin F_t, \tag{26}$$

upon which the set is updated as $S_t \leftarrow S_t \cup \{\bar{a}\}$. The procedure terminates when the ranked list is exhausted or the cardinality limit is reached, and the output is $\hat{M}_t = S_t$. Since every admission is validated against F_t, the resulting set strictly satisfies all compiled pairwise safety constraints.

Audit Rationales. After obtaining $\hat{M}_t$, structured audit rationales are constructed to support traceable inclusion and exclusion explanations. For each candidate $\bar{a} \in \bar{K}_t$, an action label is assigned as

$$\ell_t(\bar{a}) = \begin{cases} \texttt{include}, & \bar{a} \in \hat{M}_t, \\ \texttt{exclude}, & \bar{a} \notin \hat{M}_t. \end{cases} \tag{27}$$

Inclusion Rationales. For each included medication $\bar{a} \in \hat{M}_t$, inclusion evidence is retrieved from the Evidence KG by collecting edges connecting $\bar{a}$ to the visit context $\mathcal{U}_t$,

$$\text{Explain}_{\text{incl}}(\bar{a}) = \{(\bar{a}, r, u) \in E^{\leftrightarrow}_{\text{evd},t} \mid u \in \mathcal{U}_t\}, \tag{28}$$

which records context-aligned clinical associations for audit purposes only.

Exclusion Rationales. For each excluded candidate $\bar{a} \notin \hat{M}_t$, exclusion evidence is derived directly from the executable interface by identifying selected medications that conflict with $\bar{a}$,

$$\text{Explain}_{\text{excl}}(\bar{a}; \hat{M}_t) = \{\bar{b} \in \hat{M}_t \mid \{\bar{a}, \bar{b}\} \in F_t\}. \tag{29}$$

This definition guarantees decision–audit consistency for constraint-triggered exclusions, as the same feasibility constraints are used for both set construction and rationale generation. The structured audit record for visit t is thus summarized as

$$L_t = \Big(\{\text{Explain}_{\text{incl}}(\bar{a})\}_{\bar{a} \in \hat{M}_t},\ \{\text{Explain}_{\text{excl}}(\bar{a}; \hat{M}_t)\}_{\bar{a} \in \bar{K}_t \setminus \hat{M}_t}\Big), \tag{30}$$

which serves as a unified basis for subsequent review and reporting while remaining independent of the decision process itself.

4 Experiments

To systematically evaluate the proposed framework RxGuard, we investigate the following research questions:

- **RQ1:** How does RxGuard perform compared with representative baselines in terms of both predictive performance and safety?
- **RQ2:** How does each module contribute to the overall performance and safety of RxGuard?
- **RQ3:** Does RxGuard provide decision-consistent safety control and auditable rationales beyond predictive performance?

4.1 Experimental Setup

Datasets and Preprocessing. We evaluate RxGuard on two real-world longitudinal EHR datasets, MIMIC-III [11] and MIMIC-IV [10]. Following prior work [4, 5, 16, 30–32, 34, 38] for medication combination recommendation, each patient is represented as an ordered sequence of visits, where each visit consists of a set of diagnoses, a set of procedures, and a set of prescribed medications. Given the patient history and the clinical context of the target visit, the goal is to recommend the medication set for that visit. Dataset statistics are reported in Table 1. We split patients into training, validation, and test sets by patient identifier (80%/10%/10%) to avoid information leakage across splits, and tune hyperparameters on the validation set. Within each patient, visits are ordered by their recorded timestamps, and no future information beyond visit t is used when predicting medications at t. We then apply standard preprocessing to reduce sparsity and map concepts into canonical identifier spaces for consistent integration with external knowledge. Diagnoses and procedures are standardized to UMLS CUIs and deduplicated within each visit, while medications are normalized to a fixed RxNorm space using a deterministic map; if multiple raw medication identifiers map to the same canonical identifier within a visit, the canonical identifier is kept once. Unmapped diagnosis/procedure/medication codes are excluded from the standardized sets used for knowledge conditioning and constraint execution.

Table 1: Statistics of the MIMIC-III and MIMIC-IV datasets for medication combination recommendation.

Statistic	MIMIC-III	MIMIC-IV
#Patients	39,352	196,641
#Visits	50,203	463,107
Avg. visits per patient	1.28	2.36
Max visits per patient	42	98
#Unique diagnoses	6,984	28,562
#Unique procedures	2,032	14,911
#Unique medications	4,126	6,493
Avg. medications per visit	34.15	22.69
Top-10 medication coverage	0.098	0.118
Top-50 medication coverage	0.312	0.300

Knowledge Sources. We use RxNorm [26] as the canonical medication identifier space and normalize EHR medication codes to RxNorm identifiers for integrating external knowledge. For safety, we use DrugBank [13] as the source of DDI knowledge to construct the Constraint KG and compile the executable feasibility interface F_t. For audit and evidence retrieval, we use UMLS [2] as the source of external knowledge to build the Evidence KG, which is accessed only for rationale retrieval and does not affect utility calibration or set selection.

Baselines and Evaluation Metrics. We compare RxGuard with representative medication recommendation baselines from two categories. The first category includes pure EHR sequence modeling approaches that rely solely on longitudinal patient records, including RETAIN [5], Transformer [36], and DMNC [16]. The second category consists of safety-aware methods that incorporate external knowledge, including GAMENet [30], SafeDrug [38], DrugRec [32], MoleRec [39], and SSPNet [41]. We evaluate the models from two perspectives: predictive effectiveness and safety. For predictive effectiveness, we report Jaccard similarity, F1 score, and PRAUC. For safety, we measure the DDI rate.

Implementation Details. All models are trained with early stopping based on the validation set. We tune RxGuard hyperparameters separately for MIMIC-III and MIMIC-IV due to dataset-specific differences in cohort size, coding distributions, and medication-set

Table 2: Hyperparameter search spaces for RxGuard. We tune hyperparameters separately on each dataset using the validation set (effectiveness metrics with DDI rate monitored).

Hyperparameter	Search Space
Candidate size N	$\{50, 100, 150, 200\}$
Max set size $K_{\max}$	$\{20, 30, 40, 50\}$
Calibration strength β	$\{0.1, 0.2, 0.5, 1.0\}$
DDI loss weight λ	$\{0, 0.1, 0.5, 1.0\}$
Recency window Δ	$\{7, 14, 30, 60\}$
Recency decay γ	$\{0.01, 0.05, 0.1, 0.2\}$
Transformer layers L	$\{1, 2, 3, 4\}$
Hidden size d_h	$\{128, 256, 512\}$
Attention heads	$\{4, 8\}$
Learning rate	$\{1e{-}4, 3e{-}4, 1e{-}3\}$
Batch size	$\{16, 32, 64\}$
Dropout	$\{0.1, 0.2, 0.3\}$
Weight decay	$\{0, 1e{-}5, 1e{-}4\}$
Early stopping patience	10 epochs

statistics. For each dataset, we perform model selection on the validation set by optimizing predictive performance (Jaccard similarity) while monitoring DDI rate as a safety indicator. The selected configuration is then fixed and evaluated once on the held-out test set. To ensure fair comparison, baselines are tuned under the same train/validation/test split protocol and selected using the same validation-based model selection procedure whenever applicable. Table 2 summarizes the search spaces.

4.2 Overall Performance and Safety (RQ1)

Table 3 summarizes overall predictive performance and safety on MIMIC-III and MIMIC-IV. Across both datasets, RxGuard achieves the best Jaccard, F1, and PRAUC, while also obtaining the lowest DDI rate among all compared methods. Overall, the results indicate that RxGuard improves recommendation accuracy while strengthening set-level safety.

Predictive Performance. RxGuard consistently outperforms representative EHR-only baselines and recent knowledge-enhanced approaches on both datasets. The gains are observed across all three predictive metrics, suggesting that RxGuard produces more accurate medication sets rather than improvements limited to a single criterion. Notably, RxGuard remains the top-performing method on MIMIC-IV, demonstrating consistent generalization across datasets.

Safety. RxGuard reduces unsafe co-prescriptions as measured by the DDI rate compared with all baselines and knowledge-enhanced methods, and the advantage is consistent across datasets. Meanwhile, the average recommended set size remains comparable to competitive methods, indicating that the safety gains are not primarily due to recommending substantially fewer medications. Taken together, RxGuard offers a favorable accuracy and safety trade-off by enforcing safety constraints during set selection while preserving coverage of clinically relevant medications.

4.3 Ablation Study (RQ2)

We conduct an ablation study to examine how each module of RxGuard contributes to overall performance and safety. We consider four variants: *w/o Trajectory-Aware Scoring* removes longitudinal trajectory modeling and scores medications without leveraging visit-history dependencies; *w/o Local Risk Calibration* disables the constraint KG-based risk calibration and directly ranks medications by the base scores while still compiling and applying feasibility constraints; *w/o Safety-Aware Training* removes the safety-oriented training objective and optimizes the model using only the standard predictive loss; and *w/o Guarded Set Selection* removes feasibility checking at inference time and selects medications purely by score ranking under the same set-size budget.

Table 4 summarizes the results on both MIMIC-III and MIMIC-IV. The average number of recommended medications remains close across variants, suggesting that the observed differences are not primarily driven by substantial changes in set size.

Removing trajectory-aware scoring leads to a clear decline in predictive performance on both datasets, while safety changes only slightly. This indicates that longitudinal modeling is the primary contributor to recommendation accuracy, whereas safety is largely governed by downstream constraint mechanisms. In contrast, removing local risk calibration or safety-aware training increases DDI rates with moderate degradation or fluctuation in predictive metrics, implying that these two components provide complementary benefits by shaping safer rankings and encouraging safer substitutions when conflicts arise.

The most pronounced change is observed when guarded set selection is removed. In this case, predictive metrics remain comparable or slightly improve, but the DDI rate increases substantially and becomes comparable to the risk level observed in ground-truth prescriptions. This highlights that unconstrained score-based decoding alone is insufficient to prevent unsafe co-prescriptions, whereas explicit feasibility enforcement at inference time provides direct and reliable set-level safety control. Overall, the ablation results suggest that trajectory-aware scoring establishes the accuracy foundation, local risk calibration and safety-aware training refine the accuracy and safety balance, and guarded set selection serves as the key mechanism for enforcing safety constraints at decision time.

4.4 Decision Consistency and Auditability (RQ3)

This subsection evaluates whether RxGuard provides explicit decision-time safety control and verifiable audit rationales beyond predictive performance. We assess (i) constraint satisfaction of the predicted set under the compiled feasibility interface, (ii) retrievability of context-aligned evidence for included medications, and (iii) consistency between exclusion rationales and the executed constraints. For comparability, all methods are evaluated using the same global forbidden-pair set and the same UMLS-derived Evidence KG. For methods without native explanation modules, evidence is retrieved post hoc from their predicted medication sets under the same protocol.

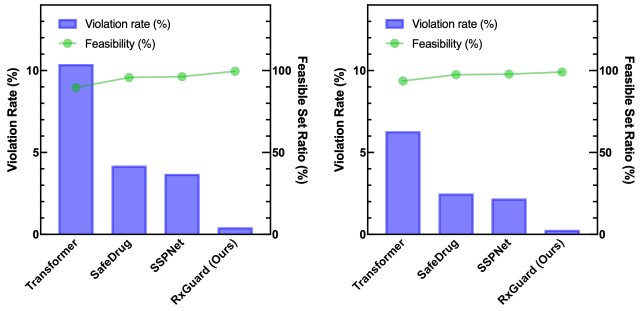
Constraint Satisfaction at Inference Time. For each visit, we evaluate whether the predicted medication set contains any forbidden DDI pair under the same executable feasibility interface used

Table 3: Overall performance and safety comparison on MIMIC-III and MIMIC-IV. Results are reported as mean(std) over 5 random seeds.

Method	MIMIC-III (GT DDI: 0.0793)					MIMIC-IV (GT DDI: 0.0738)				
	Jac↑	F1↑	PRAUC↑	DDI↓	Avg.#Med	Jac↑	F1↑	PRAUC↑	DDI↓	Avg.#Med
RETAIN	0.4687(0.0051)	0.6291(0.0058)	0.7388(0.0049)	0.0832(0.0031)	31.22(0.89)	0.3845(0.0059)	0.5382(0.0063)	0.6391(0.0055)	0.0654(0.0028)	21.08(0.82)
Transformer	0.4852(0.0048)	0.6435(0.0052)	0.7512(0.0041)	0.0805(0.0029)	32.54(0.95)	0.3989(0.0051)	0.5543(0.0058)	0.6521(0.0051)	0.0638(0.0025)	21.82(0.88)
DMNC	0.4628(0.0021)	0.6215(0.0022)	0.7334(0.0019)	0.0806(0.0020)	25.14(0.68)	0.3921(0.0023)	0.5480(0.0024)	0.6510(0.0021)	0.0629(0.0010)	19.58(0.61)
GAMENet	0.5052(0.0042)	0.6611(0.0049)	0.7605(0.0039)	0.0868(0.0033)	35.86(1.12)	0.4108(0.0048)	0.5645(0.0055)	0.6613(0.0049)	0.0785(0.0031)	24.14(0.98)
SafeDrug	0.5128(0.0039)	0.6692(0.0045)	0.7674(0.0036)	0.0665(0.0024)	27.65(0.84)	0.4193(0.0046)	0.5754(0.0051)	0.6698(0.0044)	0.0563(0.0022)	17.48(0.79)
DrugRec	0.5221(0.0035)	0.6793(0.0042)	0.7752(0.0032)	0.0615(0.0021)	28.14(0.81)	0.4324(0.0042)	0.5882(0.0048)	0.6823(0.0041)	0.0465(0.0019)	18.25(0.76)
MoleRec	0.5312(0.0032)	0.6875(0.0039)	0.7804(0.0029)	0.0732(0.0025)	31.42(0.92)	0.4462(0.0038)	0.6038(0.0045)	0.6945(0.0039)	0.0698(0.0027)	20.36(0.84)
SSPNet	0.5475(0.0028)	0.7028(0.0036)	0.7915(0.0025)	0.0797(0.0028)	33.15(0.88)	0.4876(0.0035)	0.6364(0.0042)	0.7328(0.0036)	0.0728(0.0029)	21.12(0.81)
RxGuard	**0.5534**(0.0025)	**0.7098**(0.0032)	**0.7996**(0.0021)	**0.0578**(0.0018)	30.28(0.82)	**0.4992**(0.0031)	**0.6451**(0.0038)	**0.7422**(0.0032)	**0.0445**(0.0016)	19.46(0.78)

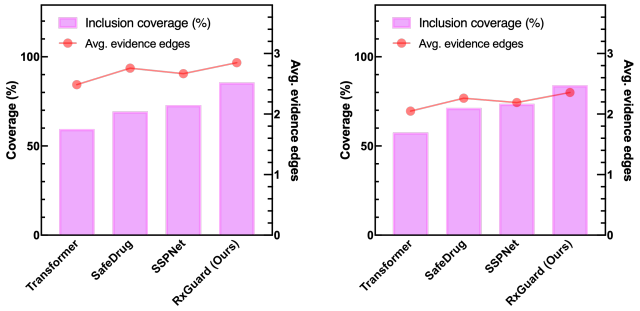
Table 4: Ablation study of RxGuard on MIMIC-III and MIMIC-IV.

Variant	MIMIC-III (GT DDI: 0.0793)					MIMIC-IV (GT DDI: 0.0738)				
	Jac↑	F1↑	PRAUC↑	DDI↓	Avg.#Med	Jac↑	F1↑	PRAUC↑	DDI↓	Avg.#Med
w/o Trajectory-Aware Scoring	0.5358(0.0036)	0.6967(0.0041)	0.7859(0.0029)	0.0589(0.0028)	29.94(0.90)	0.4811(0.0039)	0.6289(0.0046)	0.7267(0.0037)	0.0458(0.0019)	19.12(0.85)
w/o Local Risk Calibration	0.5431(0.0031)	0.7026(0.0038)	0.7920(0.0026)	0.0616(0.0021)	30.21(0.86)	0.4904(0.0034)	0.6372(0.0041)	0.7344(0.0033)	0.0489(0.0018)	19.41(0.80)
w/o Safety-Aware Training	0.5492(0.0029)	0.7068(0.0035)	0.7961(0.0024)	0.0647(0.0023)	30.62(0.88)	0.4948(0.0032)	0.6412(0.0039)	0.7378(0.0031)	0.0518(0.0020)	19.93(0.82)
w/o Guarded Set Selection	0.5560(0.0027)	0.7124(0.0033)	0.8012(0.0022)	0.0819(0.0034)	30.31(0.84)	0.5018(0.0031)	0.6478(0.0038)	0.7446(0.0030)	0.0716(0.0029)	19.51(0.79)
RxGuard (full)	**0.5534**(0.0025)	**0.7098**(0.0032)	**0.7996**(0.0021)	**0.0578**(0.0018)	30.28(0.82)	**0.4992**(0.0031)	**0.6451**(0.0038)	**0.7422**(0.0032)	**0.0445**(0.0016)	19.46(0.78)



(a) MIMIC-III (b) MIMIC-IV

Figure 2: Constraint satisfaction under the feasibility interface. We report the violation rate (percentage of visits whose predicted set contains at least one forbidden pair) and feasibility (percentage of visits with no forbidden pair) across four methods.



(a) MIMIC-III (b) MIMIC-IV

Figure 3: Evidence retrievability for included medications. We report inclusion evidence coverage (fraction of selected medications with non-empty inclusion evidence) and the average number of retrieved evidence edges per selected medication.

for all methods. Figure 2 reports the violation rate and feasibility across methods. RxGuard yields near-zero violations on both datasets, while score-based baselines exhibit non-negligible unsafe co-prescriptions, showing that explicitly enforcing constraints during decoding is necessary for reliable set-level safety.

Evidence Retrievability for Included Medications. We measure evidence retrievability using a unified external evidence source for all methods. For each method, we take its predicted medication set and retrieve context-aligned evidence from the same Evidence KG built from UMLS, then report (i) inclusion evidence coverage (the fraction of selected medications with at least one retrieved evidence item) and (ii) the average number of retrieved evidence edges per selected medication (Figure 3). This evaluation reflects how often external evidence is available for the recommended medications, rather than whether a baseline natively provides an explanation module. RxGuard achieves the highest coverage while keeping the retrieved evidence concise, and it further organizes the retrieved evidence into structured audit records for review.

Decision–Audit Consistency for Exclusion Rationales. For each excluded candidate, we verify whether the exclusion rationale corresponds to a true conflict with at least one selected medication

Table 5: Decision–audit consistency for exclusion rationales. We report the decision–audit equivalence rate, the constraint-triggered exclusion rate, and the average number of returned conflicting medications as exclusion reasons.

Metric	MIMIC-III	MIMIC-IV
Decision–audit equivalence rate (%) ↑	99.6	99.8
Constraint-triggered exclusion rate (%) ↑	92.4	88.7
Avg. #conflicts returned ↓	1.36	1.28

under the same feasibility interface used in decoding. Table 5 reports the decision–audit equivalence rate, the constraint-triggered exclusion rate, and the average number of returned conflicting medications. The results show near-perfect equivalence and concise conflict sets, supporting that exclusion rationales are derived from the executed constraints rather than generated post hoc.

Overall, RxGuard provides explicit constraint enforcement and structured audit information with decision-consistent rationales across datasets.

5 Conclusion

We proposed **RxGuard**, a decision-oriented framework for safe medication combination recommendation that shifts safety control to the inference stage. RxGuard compiles external knowledge into visit-specific decision artifacts, including a Constraint KG compiled into an executable feasibility interface for set-level safety enforcement and an Evidence KG for retrievable audit information, and then performs guarded set selection by explicitly executing feasibility constraints during decoding while producing decision-consistent inclusion evidence and constraint-aligned exclusion rationales. Experiments on MIMIC-III and MIMIC-IV show that RxGuard improves predictive effectiveness and reduces unsafe co-prescriptions compared with strong baselines, while additional analyses validate explicit constraint satisfaction, evidence retrievability, and decision–audit consistency.

Limitations and Ethical Considerations

Limitations. While RxGuard provides explicit set-level safety control and decision-consistent auditability, several limitations remain. First, the current feasibility interface primarily models pairwise interaction constraints; although this design ensures efficient and verifiable enforcement, it does not yet cover richer clinical constraints such as dosage compatibility or multi-drug conditional contraindications. Second, the quality and coverage of decision artifacts depend on the completeness of external knowledge sources. Although the framework is designed to be knowledge-source agnostic, missing or noisy relations in real-world databases may limit evidence retrievability or constraint coverage in certain cases. Third, guarded greedy decoding is an efficient approximation to the constrained set optimization problem, which may not always achieve the global optimum under complex utility landscapes, although it enables deterministic and scalable inference in practice. These limitations mainly concern extensibility and knowledge coverage rather than the core decision mechanism, and suggest directions for future improvements in richer constraint modeling, knowledge integration, and optimization strategies.

Ethical Considerations. This study uses de-identified EHR data from MIMIC-III and MIMIC-IV, accessed via PhysioNet after completing the required training and data use agreements. We comply with the corresponding requirements for data handling, storage, and use. We do not attempt any re-identification, linkage to external datasets, or sharing of patient-level records with external APIs or third-party services. The proposed system is designed for retrospective research and should not be used as a standalone clinical decision-making tool. Any real-world use would require rigorous external validation, continuous monitoring for performance disparities across patient subgroups, and appropriate clinical governance.

GenAI Disclosure

Generative AI tools were used for limited language polishing and grammar correction of non-technical text. All research ideas, methodology, experiments, and conclusions were developed and verified by the human authors. The authors take responsibility for the content of this paper.

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A Notation

Table 6 summarizes the main symbols used throughout the paper.

B Additional Details

This section summarizes implementation-level details for data processing, external knowledge construction, and evaluation protocols used in RxGuard. Unless stated otherwise, the same procedures are applied to both MIMIC-III and MIMIC-IV.

B.1 External Knowledge and Compilation

DDI Constraints. RxGuard models interaction-safety feasibility via an executable feasibility interface F_t (Eq. (17)) compiled from DrugBank DDI knowledge (Section 4.1). DrugBank interaction records are converted into a global set of forbidden medication pairs in the RxNorm-normalized identifier space. Each interaction is treated as an unordered pair and deduplicated, yielding

$$\mathcal{F} = \big\{\{a, b\} \subseteq \mathcal{V}_m \;\big|\; a \neq b,\ (a, b) \in \mathcal{E}_{\text{DDI}}\big\}. \tag{31}$$

where $\mathcal{E}_{\text{DDI}}$ is the set of DDI drug pairs extracted from DrugBank after code normalization.

For each visit t, we compile an instance-specific interface by restricting to the normalized candidate scope $\bar{K}_t = \Phi_m(K_t)$:

$$F_t = \big\{\{\bar{a}, \bar{b}\} \in \mathcal{F} \;\big|\; \bar{a} \in \bar{K}_t,\ \bar{b} \in \bar{K}_t\big\}. \tag{32}$$

All DDI-based safety evaluations (including DDI rate and the constraint violation/feasibility analysis in Figure 2) are computed using the same global forbidden-pair set $\mathcal{F}$ (restricted to the evaluated set when applicable), ensuring that decoding-time feasibility enforcement and evaluation-time safety checks are aligned and comparable across methods.

Evidence Knowledge for Audit Retrieval. RxGuard retrieves structured evidence from UMLS (Section 4.1) and represents it as a typed concept graph. Relation types used for evidence retrieval are collected into $\mathcal{R}_{\text{evd}}$ (Section 3.3). For each visit t, we build a visit-specific Evidence KG $G_{\text{evd},t}$ over $V_t = \bar{K}_t \cup \mathcal{U}_t$ (Eq. (15)) by selecting edges incident to candidate medications and connecting to visit-relevant nodes:

$$E_{\text{evd},t} = \{(x, r, y) \in E \mid x \in \bar{K}_t,\ y \in V_t,\ r \in \mathcal{R}_{\text{con}} \cup \mathcal{R}_{\text{evd}}\}, \tag{33}$$

followed by undirected closure for direction-invariant retrieval (Eq. (19)). Evidence retrieval is *read-only* and performed only after decoding: inclusion evidence is retrieved using Eq. (28) by collecting edges connecting included medications to the current visit context, and it only populates the audit record L_t (Eq. (30)). The Evidence KG is not used in computing calibrated utilities (Eq. (22)) and is not consulted in feasibility checking (Eq. (26)); therefore, audit retrieval does not affect the recommendation outcome.

B.2 Evaluation Protocol for RQ3

We evaluate decision-time constraint satisfaction, evidence retrievability, and decision–audit consistency using unified protocols for comparability across methods.

Constraint Satisfaction. For each method and visit, we check whether the predicted set $\hat{M}_t$ contains any forbidden pair under the same global forbidden-pair set $\mathcal{F}$. A visit is counted as a violation if there exists $\{a, b\} \subseteq \hat{M}_t$ with $\{a, b\} \in \mathcal{F}$, and feasible otherwise. Figure 2 reports the violation rate and feasibility over all test visits.

Evidence Retrievability. Given a predicted set $\hat{M}_t$ and visit context $\mathcal{U}_t$, we retrieve inclusion evidence for each included medication using Eq. (28) from the same UMLS-derived Evidence KG. Inclusion evidence coverage is the fraction of included medications with non-empty retrieved evidence. The average number of evidence edges is computed by averaging $|\text{Explain}_{\text{incl}}(\bar{a})|$ over included medications (Figure 3). This protocol measures evidence availability for the produced recommendations, independent of whether a baseline provides a native explanation module.

Decision–Audit Consistency for Exclusions. For RxGuard, exclusion rationales are derived from the executed interface by Eq. (29). We verify, for each excluded candidate $\bar{a} \in \bar{K}_t \setminus \hat{M}_t$, whether the returned conflict set is consistent with the compiled interface (i.e., each returned medication is selected and forms a forbidden pair with $\bar{a}$ under F_t). We report the decision–audit equivalence rate, the constraint-triggered exclusion rate (non-empty conflict sets), and the average number of returned conflicts (Table 5).

C Audit Log Example

This section gives a compact illustrative audit record showing the executed feasibility interface and the corresponding inclusion/exclusion audit items. All concepts, medications, scores, and evidence edges are placeholders used only to demonstrate the output structure; no real patient record is referenced or reconstructed.

Audit record for one visit t (illustrative).

Visit context.
$\mathcal{U}_t$ = {Hyperlipidemia, Bacterial_infection}.
Candidates.
$\bar{K}_t$ = {simvastatin, azithromycin, clarithromycin, erythromycin}.
Feasibility interface.
$F_t = \big\{\{\text{simvastatin, clarithromycin}\}, \{\text{simvastatin, erythromycin}\}\big\}$.
Decision.
$\hat{M}_t$ = {simvastatin, azithromycin}.
Inclusion evidence (stored).
- simvastatin: (simvastatin, indicated_for, Hyperlipidemia)
- azithromycin: (azithromycin, indicated_for, Bacterial_infection)

Exclusion rationales (stored).
- clarithromycin:
  $\text{Explain}_{\text{excl}}$ = {simvastatin}
  since {simvastatin, clarithromycin} $\in F_t$
- erythromycin:
  $\text{Explain}_{\text{excl}}$ = {simvastatin}
  since {simvastatin, erythromycin} $\in F_t$

Table 6: Notation summary of RxGuard.

Symbol	Definition
$\mathcal{P}$	Patient set.
p	A patient index, $p \in \mathcal{P}$.
T_p	Number of visits for patient p.
$\mathbf{X}^{(p)} = [x_1^{(p)}, \ldots, x_{T_p}^{(p)}]$	Longitudinal EHR visit sequence of patient p.
$x_t = (D_t, P_t, M_t, \tau_t)$	Visit tuple at time t: diagnoses, procedures, medications, timestamp.
$D_t \subseteq \mathcal{V}_d$	Diagnosis set at visit t (standardized as CUIs when applicable).
$P_t \subseteq \mathcal{V}_p$	Procedure set at visit t (standardized as CUIs when applicable).
$M_t \subseteq \mathcal{V}_m$	Ground-truth medication set at visit t (EHR identifier space).
τ_t	Timestamp of visit t.
$X_t = (D_t, P_t, \tau_t)$	Observable context at visit t (no medications).
$H_{t-1} = [(D_i, P_i, M_i, \tau_i)]_{i=1}^{t-1}$	Patient history available before visit t.
$\mathcal{U}_t = D_t \cup P_t$	Set-valued visit context for knowledge conditioning.
$\mathcal{V}_d, \mathcal{V}_p, \mathcal{V}_m$	Vocabularies of diagnoses, procedures, and medications, respectively.
$f_\theta : (H_{t-1}, X_t) \to \hat{M}_t$	Parameterized predictor that outputs the recommended medication set.
$\hat{M}_t \subseteq \mathcal{V}_m$	Predicted medication set at visit t.
$s_t \in \mathbb{R}^{\lvert\mathcal{V}_m\rvert}$	Base medication logits over the medication vocabulary.
$K_t = \mathrm{TopN}(s_t)$	Candidate medication set of size N selected from s_t.
N	Candidate set size.
Φ_m	Medication normalization map from EHR identifiers to canonical identifiers.
$\bar{a} = \Phi_m(a)$	Canonical identifier for medication a.
$\bar{K}_t = \Phi_m(K_t)$	Canonical candidate set at visit t.
$\bar{s}_t(\bar{a})$	Canonical base logit for $\bar{a}$ induced from s_t (Eq. (20)).
$G = (V, E, \mathcal{R})$	Global typed multi-relational knowledge graph.
$\mathcal{R}$	Relation-type set of the global KG.
$\mathcal{R}_{\mathrm{con}}$	Constraint relation types (DDI constraints between medications).
$\mathcal{R}_{\mathrm{evd}}$	Evidence relation types used for audit retrieval.
$V_t = \bar{K}_t \cup \mathcal{U}_t$	Visit-specific node scope for decision compilation.
$G_{\mathrm{con},t} = (V_t, E_{\mathrm{con},t})$	Constraint KG induced on visit t.
$E_{\mathrm{con},t}$	Constraint edges among $\bar{K}_t$ (Eq. (16)).
F_t	Executable feasibility interface (forbidden unordered pairs) compiled from $G_{\mathrm{con},t}$ (Eq. (17)).
$G_{\mathrm{evd},t} = (V_t, E^{\leftrightarrow}_{\mathrm{evd},t})$	Evidence KG for visit t (Eq. (19)).
$E^{\leftrightarrow}_{\mathrm{evd},t}$	Undirected closure of evidence edges for retrieval.
$\mathcal{N}_t(\bar{a})$	Neighbor set of $\bar{a}$ in the constraint-induced candidate graph.
$r_t(\bar{a})$	Conflict risk for $\bar{a}$ computed from neighbors (Eq. (21)).
$\tilde{s}_t(\bar{a})$	KG-calibrated utility for $\bar{a}$ (Eq. (22)).
β	Calibration strength in risk inhibition.
λ	DDI loss weight in the training objective.
$K_{\max}$	Maximum predicted set size in constrained decoding.
$\mathrm{Explain}_{\mathrm{incl}}(\bar{a})$	Inclusion evidence for $\bar{a}$ retrieved from $G_{\mathrm{evd},t}$.
$\mathrm{Explain}_{\mathrm{excl}}(\bar{a}; \hat{M}_t)$	Exclusion rationale for $\bar{a}$ derived from F_t and $\hat{M}_t$.
L_t	Structured audit record at visit t.